

SpecFirst React A11y

Phase 5 Documentation: Test Generation

How the locked accessibility contract becomes executable Playwright and axe-core tests

Phase 5 is the bridge between the frozen contract and baseline-red verification. It reads `lockedSpec.json`, generates deterministic test code, records the generated artifacts, and then stops. It does not run tests or involve Bob.

Prepared for the IBM Bob Hackathon workflow

1. Phase 5 in one sentence

Phase 5 turns the frozen lockedSpec.json contract from Phase 4 into deterministic Playwright and axe-core tests, without running those tests and without allowing Bob to modify the contract or test file.

In the pipeline, Phase 5 sits here:

Phase 4: Freeze spec -> Phase 5: Generate tests -> Phase 6: Run baseline-red verification

2. Purpose

- Make the locked contract executable.
- Generate one named test for each automated check in lockedSpec.json.
- Preserve traceability from test failures back to locked check IDs.
- Create a stable test artifact that Bob must not edit later.
- Prepare Phase 6 to prove the current broken component fails before remediation.

The core principle is: SpecFirst defines and writes the tests before Bob patches the component.

3. Inputs

Input	Purpose
lockedSpec.json	The source of truth for all generated tests. Must have status locked, integrity hash, checks, component bindings, component scope, and test harness hints.
Component source path	Used by the generated harness/import. Phase 5 does not modify the component.
Test harness hints	Provide label, demo options, and onChange/state requirements for rendering the component in a testable environment.
Component scope selector	Defines the root used for scoped axe-core scans, usually a demo wrapper data attribute.

4. Outputs

Output	Purpose
tests/a11y/.spec.ts	Generated Playwright test file. Main Phase 5 output.
testGenerationResult.json	Run-local metadata about generated tests, check IDs, locked spec hash, test file path, and baseline command.
testFileHash	Hash of generated test file. Used later to verify Bob did not alter the tests.
Optional generated harness file	If the project does not already expose a stable demo route, generate a wrapper/harness for rendering the component with deterministic props.

5. Scope boundaries

Phase 5 should do:

- Validate lockedSpec.json and refuse unlocked, skipped, or failed specs.

- Generate Playwright tests from locked checks and strategy templates.
- Generate or reference a test harness that renders the component inside a scope wrapper.
- Include locked spec path/hash in the generated test file header.
- Write testGenerationResult.json and compute a test file hash.

Phase 5 should not do:

- Run Playwright tests. That is Phase 6.
- Call Bob. Bob appears later when remediation starts.
- Patch the React component.
- Modify lockedSpec.json.
- Decide pass/fail or make accessibility compliance claims.

6. Generated test structure

Each automated check in lockedSpec.json becomes one named test. The test name must include the check ID so failures map directly back to the locked contract.

Generated file header should include:

```
// AUTO-GENERATED BY SPECFIRST // Source locked spec: specfirst/runs//lockedSpec.json // Locked spec hash: sha256: //
Bob must not edit this file.
```

7. Strategy-to-template mapping

Test strategy	Generated behavior
role-name-query	Use Playwright role locator with expected accessible name for the combobox.
role-query	Query expected roles such as combobox, listbox, and option.
attribute-toggle	Assert an attribute before and after interaction, usually aria-expanded.
attribute-check	Assert a structural/state attribute such as aria-controls or aria-selected.
keyboard-open	Focus trigger, press ArrowDown, expect popup and active option semantics.
keyboard-select	Open popup, activate option, press Enter, expect selection behavior.
keyboard-close	Open popup, press Escape, expect popup closed and focus returned or retained.
axe-scan	Run WCAG-tagged axe-core scan scoped to component root in collapsed and expanded states.

8. MVP checks to generate

- accessible-name
- combobox-role
- expanded-state
- popup-associated
- popup-listbox
- option-roles
- selected-state
- keyboard-arrow-down-opens
- keyboard-enter-selects
- keyboard-escape-closes
- axe-scan

9. Harness requirements

The generated tests need a deterministic way to render the component. The locked spec gives Phase 5 enough information to create or reference that harness: component name, component path, label prop

need, options collection need, onChange spy/state need, and component scope selector.

Recommended MVP harness behavior:

- Render the target component inside a wrapper matching componentScope.selector, for example data-specfirst-root="SelectOnlyCombobox".
- Pass label="Choose fruit" or the label inferred by the locked spec.
- Pass deterministic options, usually Apple and Banana for the MVP fixture.
- Manage a local value state so selection tests can observe changes.
- Expose the harness through a stable route or a generated test-only page.

10. Recommended generated artifact: testGenerationResult.json

The result artifact should make the generation step auditable and prepare Phase 6.

```
{ "schemaVersion": "1.0.0", "status": "generated", "lockedSpecPath": "specfirst/runs//lockedSpec.json", "lockedSpecHash":  
"sha256:", "testFilePath": "tests/a11y/SelectOnlyCombobox.spec.ts", "testFileHash": "sha256:", "generatedCheckIds":  
["accessible-name", "combobox-role", "..."], "testCount": 11, "nextPhase": { "canContinue": true, "baselineTestCommand":  
"npx playwright test tests/a11y/SelectOnlyCombobox.spec.ts" } }
```

11. How each MVP check should behave

Check	Generated assertion intent
accessible-name	Expect a combobox with a stable accessible name derived from the locked spec binding.
combobox-role	Expect the trigger/control to expose role=combobox.
expanded-state	Assert aria-expanded is false collapsed, true open, false after close.
popup-associated	Assert aria-controls exists and points to a listbox element.
popup-listbox	Open popup and expect getByRole('listbox') to resolve.
option-roles	Open popup and expect mapped options to expose role=option.
selected-state	After selection, expect aria-selected=true on selected option and false on others.
keyboard-arrow-down-opens	Focus trigger, press ArrowDown, expect popup open and first option active via aria-activedescendant strategy.
keyboard-enter-selects	With active option, press Enter and expect selection plus popup close.
keyboard-escape-closes	Open popup, press Escape, expect popup closed and focus on/returned to trigger.
axe-scan	Run WCAG-tagged axe-core scan scoped to component root in collapsed and expanded states.

12. Important correctness notes

- The generated keyboard tests must honor the locked focusManagementStrategy. For the MVP, DOM focus remains on the combobox and the active option is represented through aria-activedescendant.
- The generated axe test should use WCAG tags from the locked spec and scope the scan to componentScope.selector.
- The generated test should be strict enough to fail against the broken baseline before Bob patches the component.
- The test file should never be edited by Bob. Its hash must be preserved for later verification.
- Phase 5 should fail on unsupported testStrategy values rather than silently skipping them.

13. Status model

Status	Meaning
generated	Tests were generated successfully from a locked spec.
skipped	Upstream spec did not allow continuation.
failed	Spec was invalid for test generation or an unsupported strategy was encountered.

14. Failure cases

- lockedSpec.json is missing or status is not locked.

- integrity.hash is missing.
- nextPhase.canContinue is false.
- checks array is empty.
- A check has an unsupported testStrategy.
- componentScope.selector is missing.
- testHarnessHints are insufficient to render the component.
- test file cannot be written or hashed.

15. Test plan for Phase 5

- Golden fixture: supported select-only combobox generates 11 tests.
- Each generated test name includes the locked check ID.
- Generated file header contains locked spec path and hash.
- testGenerationResult.json contains generatedCheckIds, testCount, testFileHash, and baseline command.
- Unsupported test strategy returns failed.
- Skipped/failed locked specs do not produce tests.
- Changing lockedSpec hash or check list changes the generation result/hash.
- Generated tests are deterministic across repeated runs with the same locked spec.

16. Definition of done

- Phase 5 reads lockedSpec.json and validates it before generation.
- All 11 MVP checks are converted into Playwright/axe test cases.
- A stable component harness strategy is defined or generated.
- Generated tests are traceable to locked check IDs.
- Generated tests are not run in Phase 5.
- Bob is not invoked and must not edit the test file.
- testGenerationResult.json and test file hash are produced.
- Phase 6 can run the baseline command directly.

End of Phase 5 documentation.